Lecture 7

Laplace Transform

7.1 Laplace Transform

ラプラス変換

7.1 Laplace Transform

Laplace transforms are invaluable for any engineer' s mathematical toolbox as they make **solving linear Ordinary Differential Equations (ODEs)† (常微分方程式)** and related initial value problems

The process of solving an ODE using the **Laplace transform** method consists of three steps:

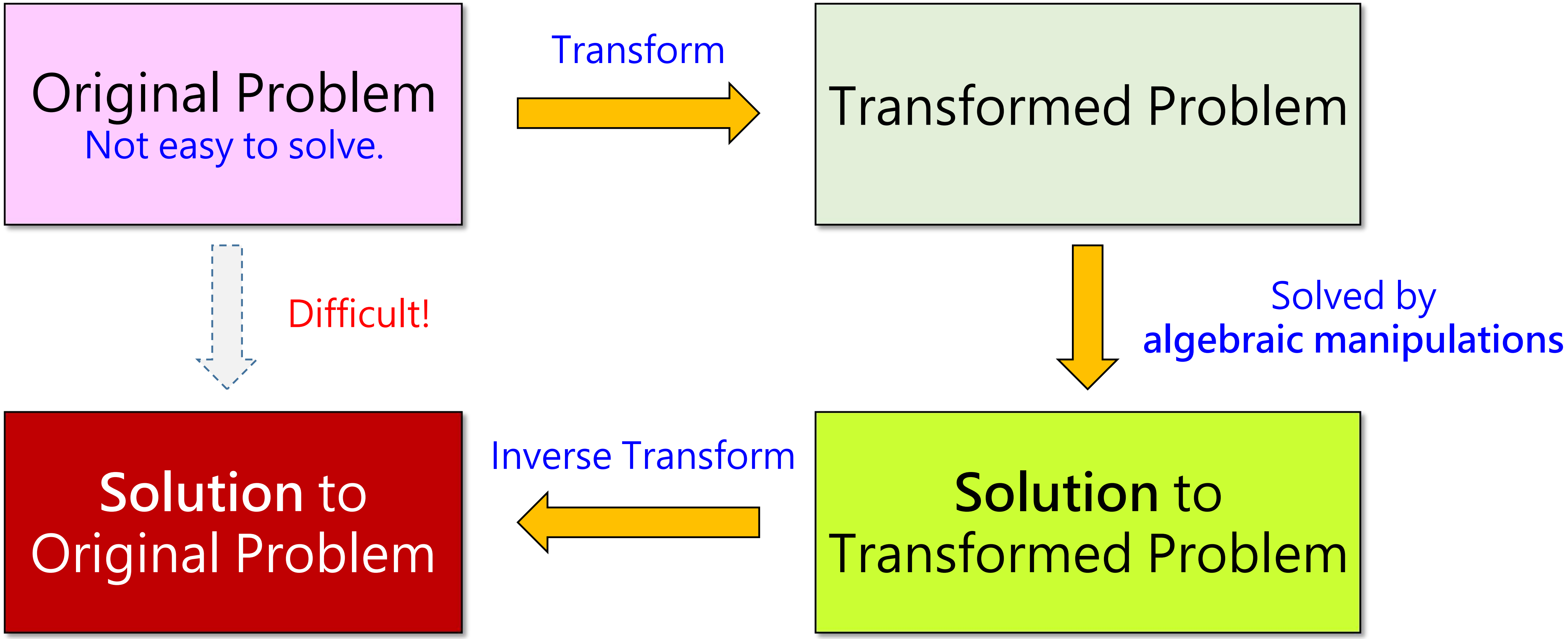
Step 1. The given **ODE** is **transformed into** an **algebraic equation (代数方程式)**, called the **subsidiary equation (補助方程式)**.

Step 2. The **subsidiary equation** is **solved by purely algebraic manipulations**. (補助方程式は、純粋に代数的操作によって解かれます。)

Step 3. **The solution in Step 2 is transformed back**, resulting in the solution of the given problem.

†ODEs: https://en.wikipedia.org/wiki/Ordinary_differential_equation

7.1 Laplace Transform



7.1 Laplace Transform

Suppose that $f(t)$ is defined for all $t \geq 0$.

The **Laplace transform** of f is the function

$$L(s) = \mathcal{L}(f)(s) = \int_0^{\infty} f(t) e^{-st} dt \qquad (7.1)$$

The functions f, g,... are defined for $t \geq 0$,

and their transforms are defined on the s-axis.

Notice:

Original functions depend on t and **their transforms on** s,

keep this in mind!

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Furthermore, the given function $f(t)$ in (7.1) is called the

Inverse Laplace Transform of $L(s)$ and is denoted by

$\mathcal{L}^{-1}(\mathcal{L}(f)(s))$; we shall write

$$f(t) = \mathcal{L}^{-1}(\mathcal{L}(f)(s)) = \mathcal{L}^{-1}(L(s)) \tag{7.2}$$

Note that (7.1) and (7.2) together imply that we have a new **transform pair**

$$\boxed{\begin{aligned} L(s) &= \mathcal{L}(f(t))(s) \\ f(t) &= \mathcal{L}^{-1}(L(s))(t) \end{aligned}}$$

Laplace Transform Pair

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We say that f is of **exponential order** if there exist **nonnegative numbers** u and M such that

$$|f(t)| \leq Me^{ut} \text{ for all } t \geq 0 \tag{7.3}$$

Theorem 7.1 Existence of Laplace Transform

Suppose that f is piecewise continuous on the interval $[0, \infty)$ and of exponential order with $|f(t)| \leq Me^{ut}$ for all $t \geq 0$. Then $\mathcal{L}(f)(s)$ exists for all $s > u$.

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Example 7.1 Find $\mathfrak{L}(1), \mathfrak{L}(t), \mathfrak{L}(e^{at})$

Solution: We compute these Laplace Transforms by using (7.1). We have

$$\mathfrak{L}(1)(s) = \int_0^\infty 1 \cdot e^{-st} dt = -\frac{1}{s} e^{-st}\Big|_0^\infty = \frac{1}{s} \qquad s > 0$$

$$\int_0^\infty 1 \cdot e^{-st} dt = \lim_{T\to\infty} \int_0^T e^{-st} dt = \lim_{T\to\infty} \left[-\frac{1}{s} e^{-st}\right]_0^T = \lim_{T\to\infty} \left[-\frac{1}{s} e^{-sT} + \frac{1}{s} e^0\right] = \frac{1}{s} \qquad s > 0$$

$$\mathfrak{L}(t)(s) = \int_0^\infty t \cdot e^{-st} dt = \lim_{T\to\infty} \int_0^T t \cdot e^{-st} dt = \lim_{T\to\infty} \left[-\frac{t}{s} e^{-st} - \frac{1}{s^2} e^{-st}\right]_0^T = \frac{1}{s^2} \qquad s > 0$$

Integrating by parts

$$\mathfrak{L}(e^{at})(s) = \int_0^\infty e^{at} \cdot e^{-st} dt = \int_0^\infty e^{-(s-a)t} dt = -\frac{1}{s-a} e^{-(s-a)t}\Big|_0^\infty = \frac{1}{s-a} \qquad s > a$$

Theorem 7.2 Linearity of Laplace Transform

If f and g are functions and a and b are real numbers, then

$$\mathcal{L}(af + bg) = a\mathcal{L}(f) + b\mathcal{L}(g) \qquad (7.4)$$

Proof:

$$\mathcal{L}(af(t) + bg(t)) = \int_0^\infty e^{-st}[af(t) + bg(t)]dt$$

$$= a\int_0^\infty e^{-st} f(t)dt + b\int_0^\infty e^{-st} g(t)dt$$

$$= a\mathcal{L}(f(t)) + b\mathcal{L}(g(t))$$

$$\mathcal{L}^{-1}(a\mathcal{L}(f) + b\mathcal{L}(g)) = a\mathcal{L}^{-1}(\mathcal{L}(f)) + b\mathcal{L}^{-1}(\mathcal{L}(g)) \qquad (7.5)$$

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Example 7.2 **Application of Theorem 7.2: Hyperbolic Functions**

Find the Laplace Transforms of $\cosh\, at$ and $\sinh\, at$

Solution:

$$\mathcal{L}(\cosh at)(s) = \mathcal{L}\left(\frac{e^{at}+e^{-at}}{2}\right)(s) = \frac{1}{2}\left[\mathcal{L}(e^{at})(s) + \mathcal{L}(e^{-at})(s)\right] = \frac{1}{2}\left(\frac{1}{s-a}+\frac{1}{s+a}\right) = \frac{s}{s^2-a^2}$$

$$\mathcal{L}(\sinh at)(s) = \mathcal{L}\left(\frac{e^{at}-e^{-at}}{2}\right)(s) = \frac{1}{2}\left[\mathcal{L}(e^{at})(s) - \mathcal{L}(e^{-at})(s)\right] = \frac{1}{2}\left(\frac{1}{s-a}-\frac{1}{s+a}\right) = \frac{a}{s^2-a^2}$$

- Hyperbolic sine (双曲線正弦):
  $$\sinh x = \frac{e^x - e^{-x}}{2} = \frac{e^{2x}-1}{2e^x} = \frac{1-e^{-2x}}{2e^{-x}}.$$
- Hyperbolic cosine (双曲線余弦):
  $$\cosh x = \frac{e^x + e^{-x}}{2} = \frac{e^{2x}+1}{2e^x} = \frac{1+e^{-2x}}{2e^{-x}}.$$
- Hyperbolic tangent (双曲線正接):
  $$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \\ = \frac{e^{2x}-1}{e^{2x}+1} = \frac{1-e^{-2x}}{1+e^{-2x}}.$$

Example 7.3 **Application of Theorem 7.2: Cosine and Sine**

Find the Laplace Transforms of $\cos kt$ and $\sin kt$

Solution:

According to Euler's Identity $e^{ikt} = \cos kt + i\sin kt$

$$\mathcal{L}(\cos kt) + i\mathcal{L}(\sin kt) = \int_0^\infty e^{-st}(\cos kt + i\sin kt)dt = \int_0^\infty e^{-st}e^{ikt}dt$$

$$= \int_0^\infty e^{-(s-ik)t}dt$$

$$= -\frac{1}{s-ik}e^{-(s-ik)t}\Big|_0^\infty$$

$$= \frac{1}{s-ik}$$

$$= \frac{s+ik}{s^2+k^2}$$

$$= \frac{s}{s^2+k^2} + i\frac{k}{s^2+k^2}$$

Equating real and imaginary parts, we have

$$\mathcal{L}(\cos kt) = \frac{s}{s^2+k^2} \qquad s > 0$$

$$\mathcal{L}(\sin kt) = \frac{k}{s^2+k^2} \qquad s > 0$$

7.1 Laplace Transform

Table of Laplace Transform. We shall see that from these almost all the others can be obtained by the use of the general properties of the Laplace transform.

	$f(t), \quad t \geq 0$	$L(s) = \mathcal{L}(f)(s) = \int_0^\infty f(t)e^{-st}\,dt,$
1.	1	$\frac{1}{s}, \quad s > 0$
2.	$t^n, \quad n = 1, 2, \ldots$	$\frac{n!}{s^{n+1}}, \quad s > 0$
3.	$t^a \quad (a > -1)$	$\frac{\Gamma(a+1)}{s^{a+1}}, \quad s > 0$
4.	e^{at}	$\frac{1}{s-a}, \quad s > a$
5.	$t^n e^{at}$	$\frac{n!}{(s-a)^{n+1}}, \quad s > a$
6.	$\frac{e^{at}-e^{bt}}{a-b}$	$\frac{1}{(s-a)(s-b)}, \quad s > \max(a,b)$
7.	$\frac{ae^{at}-be^{bt}}{a-b}$	$\frac{s}{(s-a)(s-b)}, \quad s > \max(a,b)$
8.	$\sin kt$	$\frac{k}{s^2+k^2}, \quad s > 0$
9.	$\cos kt$	$\frac{s}{s^2+k^2}, \quad s > 0$

⋮

Nakhlé H. Asmar, Appendix B, *Partial Differential Equations with Fourier Series and Boundary Value Problems 2nd Edition*, 2004

Theorem 7.3 Shifting on the s-axis

Suppose that f is of exponential order. Let a be a **real number** and u be **nonnegative number** as in (7.3). For $s > u + a$, we have

$$\mathfrak{L}[e^{at} f(t)](s) = L(s-a) \tag{7.6}$$

If we take the inverse on both sides of (7.6).

$$e^{at} f(t) = \mathfrak{L}^{-1}[L(s-a)\ ] \tag{7.7}$$

Proof

For $s > u + a$

$$\mathfrak{L}[e^{at} f(t)] = \int_0^\infty [e^{at} f(t)] e^{-st} dt = \int_0^\infty e^{at} e^{-st} f(t) dt = \int_0^\infty e^{-(s-a)t} f(t) dt = L(s-a)$$

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Example 7.4

Known that

$$\mathfrak{L}[e^{at}\cos kt](s) = \frac{s-a}{(s-a)^2+k^2} \quad s > a \quad \text{and} \quad \mathfrak{L}[e^{at}\sin kt](s) = \frac{k}{(s-a)^2+k^2} \quad s > a$$

Find the inverse transform of $\mathfrak{L}[f](s) = \dfrac{3s-137}{s^2+2s+401}$

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Solution:

$$f(t) = \mathcal{L}^{-1}\left[\frac{3s-137}{s^2+2s+401}\right] = \mathcal{L}^{-1}\left[\frac{3s+3-140}{s^2+2s+1+400}\right] = \mathcal{L}^{-1}\left[\frac{3(s+1)-140}{(s+1)^2+400}\right]$$

$$= \mathcal{L}^{-1}\left[\frac{3(s+1)}{(s+1)^2+400} - \frac{7\cdot 20}{(s+1)^2+400}\right]$$

$$= 3\mathcal{L}^{-1}\left[\frac{s+1}{(s+1)^2+20^2}\right] - 7\mathcal{L}^{-1}\left[\frac{20}{(s+1)^2+20^2}\right]$$

By (7.5)

Taking $a = -1$, $k = 20$, we have

$$\mathcal{L}^{-1}\left[\frac{s-(-1)}{(s-(-1))^2+20^2}\right] = e^{-t}\cos 20t \qquad \mathcal{L}^{-1}\left[\frac{20}{(s-(-1))^2+20^2}\right] = e^{-t}\sin 20t \qquad s > a$$

$$f(t) = 3e^{-t}\cos 20t - 7e^{-t}\sin 20t$$

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Example 7.5 **Inverse Laplace Transforms**

(a) Evaluate $\mathcal{L}^{-1}\left(\frac{2}{4+(s-1)^2}\right)$ (b) Evaluate $\mathcal{L}^{-1}\left(\frac{1}{s^2+2s+3}\right)$

Solution: (a) From the Table of Laplace Transforms, we know

$$\mathcal{L}(e^{at}\sin kt) = \frac{k}{(s-a)^2+k^2} \qquad s > a$$

Consider that $a = 1$ and $k = 2$, we have

$$e^{t}\sin 2t = \mathcal{L}^{-1}\left(\frac{2}{2^2+(s-1)^2}\right) = \mathcal{L}^{-1}\left(\frac{2}{4+(s-1)^2}\right) \qquad s > 1$$

(b) $$\frac{1}{s^2+2s+3} = \frac{1}{(s+1)^2+(\sqrt{2})^2} = \frac{1}{\sqrt{2}}\frac{\sqrt{2}}{(s+1)^2+(\sqrt{2})^2}$$

Consider that $a =- 1$ and $k = \sqrt{2}$, we have

$$e^{-t}\sin\sqrt{2}t = \mathcal{L}^{-1}\left(\frac{\sqrt{2}}{(\sqrt{2})^2+(s+1)^2}\right) = \sqrt{2}\mathcal{L}^{-1}\left(\frac{1}{s^2+2s+3}\right) \Rightarrow \mathcal{L}^{-1}\left(\frac{1}{s^2+2s+3}\right) = \frac{1}{\sqrt{2}}e^{-t}\sin\sqrt{2}t \qquad s > a$$

Uniqueness of Laplace Transform.

If the **Laplace Transform of a given function exists**, **it is uniquely determined**.

Conversely, it can be shown that if two functions (both defined on the positive real axis) have the same transform, these functions cannot differ over an interval of positive length, although they may differ at isolated points.

Hence we may say that the inverse of a given Laplace Transform is essentially unique.

In particular, if two *continuous* functions have the same transform, they are completely identical.

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Theorem 7.4 Laplace Transform of Derivatives

(i) Suppose that f is continuous on $[0, \infty)$ and of exponential order as in (7.3).

Suppose further that f' is piecewise continuous on $[0, \infty)$ and of exponential order. Then

$$\mathcal{L}(f') = s\mathcal{L}(f) - f(0) \tag{7.8}$$

(ii) More generally, if $f, f', \ldots, f^{(n-1)}$ are continuous on $[0, \infty)$ and of exponential order as in (7.3), and $f^{(n)}$ is piecewise continuous on $[0, \infty)$ and of exponential order, then

$$\mathcal{L}(f^{(n)}) = s^n\mathcal{L}(f) - s^{n-1}f(0) - s^{n-2}f'(0) - \ldots - f^{(n-1)}\mathcal{L}(0) \tag{7.9}$$

When $n = 2$, (7.9) gives

$$\mathcal{L}(f'') = s^2\mathcal{L}(f) - sf(0) - f'(0) \tag{7.10}$$

Theorem 7.5 Derivative of Laplace Transform

(i) Suppose $f(t)$ is piecewise continuous and of exponential order. Then

$$\mathcal{L}(tf(t))(s) = -\frac{d}{ds}\mathcal{L}(f)(s) \tag{7.11}$$

(ii) In general, if $f(t)$ is piecewise continuous and of exponential order, then

$$\mathcal{L}(t^n f(t))(s) = (-1)^n \frac{d^n}{ds^n}\mathcal{L}(f)(s) \tag{7.12}$$

Example 7.6 **Derivative of Laplace Transform**

(a) Evaluate $\mathcal{L}(t\sin 2t)(s)$

(b) Evaluate $\mathcal{L}(t^2\sin t)(s)$

Solutions

(a) By using (7.11) in the Theorem 7.5 and Table of the Laplace Transform,

$$\mathcal{L}(t\sin 2t) = -\frac{d}{ds}\mathcal{L}(\sin 2t) = -\frac{d}{ds}\left[\frac{2}{s^2+4}\right] = \frac{4s}{(s^2+4)^2} \qquad \mathcal{L}(\sin kt)(s) = \frac{k}{s^2+k^2} \quad s>0$$

(b) By using (7.12) in the Theorem 7.5 and Table of the Laplace Transform,

$$\mathcal{L}(t^2\sin t) = (-1)^2\frac{d^2}{ds^2}\mathcal{L}(\sin t) = \frac{d^2}{ds^2}\left[\frac{1}{s^2+1}\right] = \frac{2(3s^2-1)}{(s^2+1)^3} \qquad \mathcal{L}(\sin kt)(s) = \frac{k}{s^2+k^2} \quad s>0$$

Review for Lecture 7

- Laplace Transform and Inverse Laplace Transform
- Linearity of Laplace Transform
- Laplace Transform of Derivatives
- Derivative of Laplace Transform

Assignment

Please Check https://web-ext.u-aizu.ac.jp/~xiangli/teaching/MA05/index.html

References

[1] Nakhlé H. Asmar, *Partial Differential Equations with Fourier Series and Boundary Value Problems 2nd Edition*, 2004

[2] Wikipedia